

# Where the project stands

Goal: rewrite NASA GISS's ROCKE-3D (Fortran GCM) in Python/JAX, checked against the real Fortran, for GPU use. Configuration P2SAoM40 (72x46 grid, 40 layers, dynamic ocean). Every card says what is met and what is not.

## NOT MET

### One-day criterion (nov26, 54 steps)

ACCEPTANCE s9: never beyond 2x the largest real-member distance at every step. QCL at step 1 is 2.84x (one bistable cell, D197). Steps  $\geq 3$ : worst ratio 1.12. 54 steps finite after the D199+D202 land fixes. D197, D203.

## MET / PARTLY MET

### One coupled step vs real Fortran (C2, step 0)

nov26 MET with one named column (25,16); dec01 and jan01 PARTLY MET (worst EGCM 9.9e-9 and 7.6e-9 of scale). Needs the libimf callback. D191.

## 18.8 s

### Steady step on CPU, was 44.3 s

99 jit executions (was 262), eager dispatches 490 (was ~15,700). Only 1.2-1.4x faster than the NumPy chain on the same cores. HYBRID step. D187, D191.

## 558 / 558

### C1: bitwise vs our NumPy chain

Step 0 on nov26, dec01, jan01 (category A). On the day: bitwise steps 0-19 (the NumPy reference aborts at step 20). D191, D203.

## 3,056 passed

### Last batch regression (a89e117)

6 skipped, 1 known failure (export artifact: test asserts a git HEAD; fixed afterwards, not re-run). README\_START\_HERE handoff.

## No GPU result

### GPU runner built, validated on CPU only

A100 probe: launch-bound for small sequential ops; device exp/sin/pow differ from NumPy by 1 ulp in 6-12% of values, so never bitwise on GPU. D198, gpu/DISCOVER\_RUN.md.

This is a HYBRID port, not end-to-end JAX: radiation is the original Fortran (served) or replayed from a record; libimf math, QUS, pole columns and the OADVT2 pre-pass are host callbacks; Ent exports, land forcing and tile radiation columns are recorded inputs (D191 s4). SOCRATES is never ported or modified.

# Status by area, and what is still outstanding

Green = verified success, yellow = in progress or partial, red = broken or not met, grey = not started. Violet marks outstanding items and their next action.

## Status by area (2026-10-08)

-  **Component ports vs the Fortran (Track B)**  
validated rung by rung, D4-D145
-  **One coupled step vs the real Fortran (C2)**  
step 0 MET on nov26, dec01, jan01 (D199); hybrid
-  **JAX step equals the NumPy chain (C1)**  
bitwise: 559/559 arrays at step 0; nov26 steps 0-19
-  **Regression suite**  
3,059 passed, 6 skipped, 0 failed (fd53982: gusti fix, D203)
-  **Device-resident assembled step**  
99 jit executions, 18.8 s/step; hybrid, not end-to-end JAX
-  **Radiation (SOCRATES)**  
replayed or Fortran callback; accepted exception, not ported
-  **GPU assembled step**  
runs on the A100 (step 0 OK) but NaN from step 1 (surface stage); cause unknown
-  **One model day vs 5 real members (ACCEPTANCE s9)**  
NOT MET: QCL step 1 = 2.84x (D203); all 54 steps finite after the land fixes
-  **Daily lake update at the day boundary**  
missing (D196): lake-tile mismatches at step 48
-  **Months-long run and its criterion**  
not started; criterion not defined

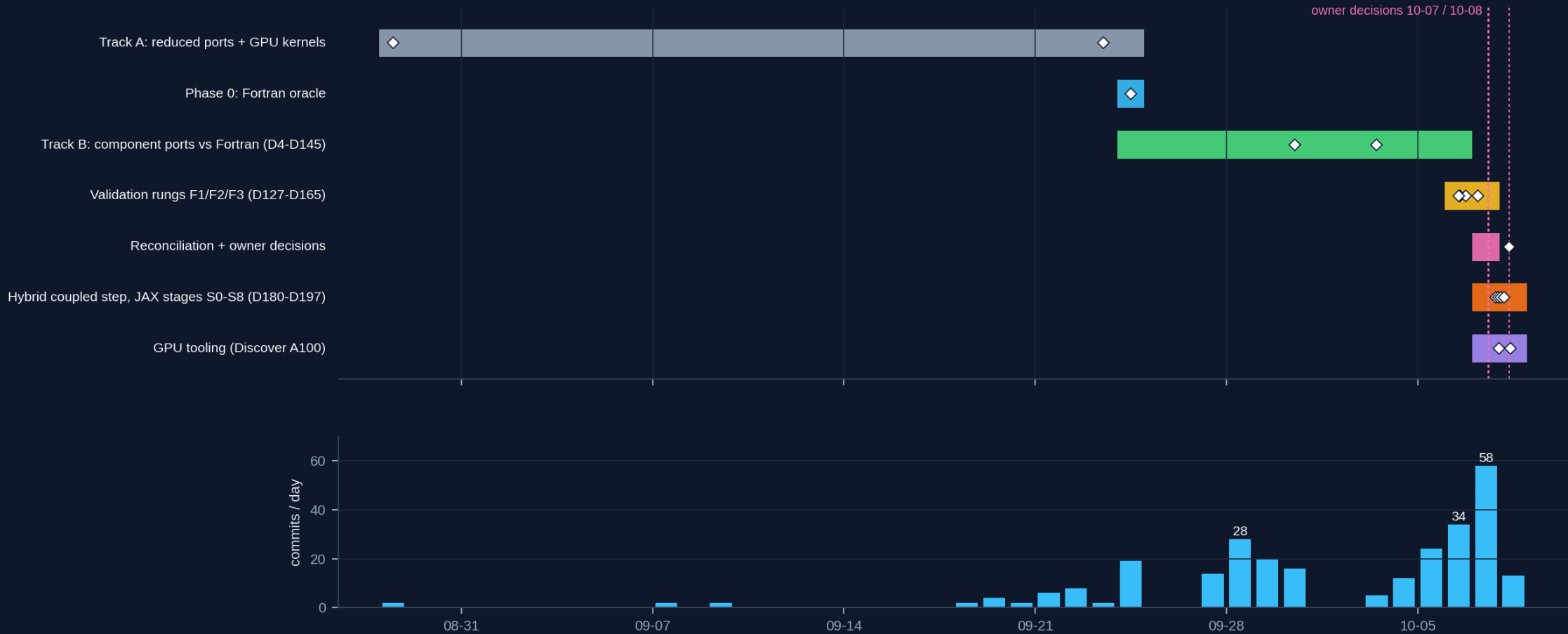
## Outstanding items and their next action

-  **GPU NaN at step 1 (surface stage)**  
owner: JAX\_DEBUG\_NANS=1 run on Discover; CPU control with the simplifier on is finite
-  **Day-boundary lake update**  
owner: record the boundary values (approximation) or port daily\_LAKE (~500 lines)
-  **QCL step 1 exceedance (2.84x)**  
one bistable cell (D197); owner may relax s9 later (dated, both statuses)
-  **Free-running comparison**  
compute Ent and land forcing instead of replaying the real records
-  **Radiation and SOCRATES**  
135-295 h port estimate; only if the owner lifts the rule (D1-D6)
-  **Multi-month criterion**  
define it; use the 8-member JAN1950 leave-one-out scoring
-  **Nit schedule and C1 beyond step 19**  
port the Fortran gdtm loop (D202 prototype); NumPy reference aborts at step 20
-  **Housekeeping**  
owner: close extra sessions; decide where ANN4099 lives; push when batches pass

**Honest status: the single step matches the real Fortran at step 0 on three dates, but the step is a hybrid and the one-day statistical test is not passed yet.**

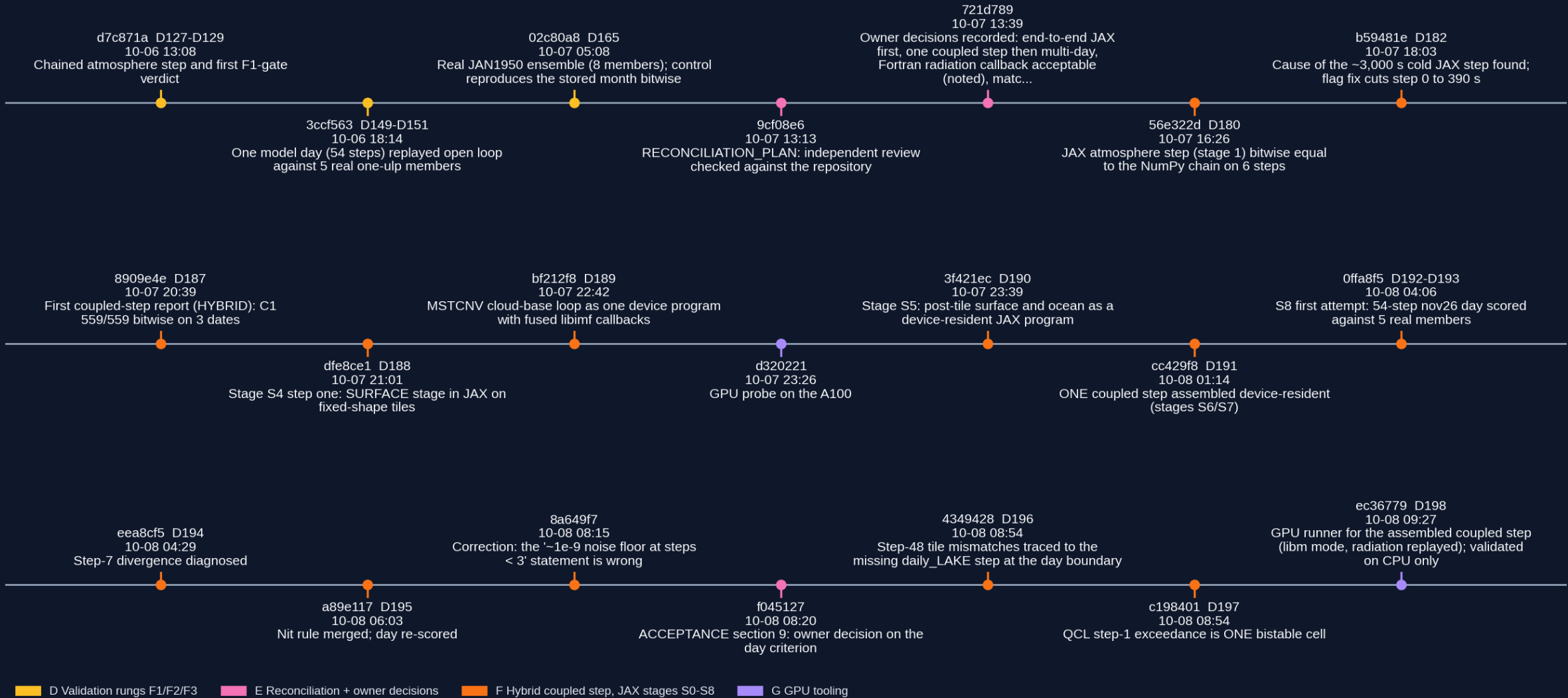
# Six weeks, 273 commits: from reduced ports to a hybrid JAX coupled step

ROCKE-3D -> JAX: chronology, 2026-08-28 to 2026-10-08 (273 commits; diamonds = milestones listed in PROJECT\_CHRONOLOGY.md)



# From the F1 gate to the one-day verdict: 2026-10-06 to 2026-10-08

Zoom: 2026-10-06 to 2026-10-08, events in commit order (not to time scale); date/time = commit time; colors = phase



Sources: git log; ledger D118-D198 (FULL\_FIDELITY\_DELTAS.md); ACCEPTANCE s8-9. Owner decisions: 2026-10-07 13:39 / 14:19 / 14:25 and 2026-10-08 08:20.

# Seven phases, seven lessons

| Phase                                           | Dates          | Key commits / ledger                      | What was learned                                                                                                                                                                                       |
|-------------------------------------------------|----------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A Track A: reduced component ports, GPU kernels | 08-28 to 09-24 | c2991dd, 5cbe7f8, abd805b                 | Reduced drivers ran fast (kernels 2.3-6.3x on GPU; fused chain 4.2x) but were not faithful; Track A recommended against a full port, later reversed by the owner.                                      |
| B Phase 0: Fortran oracle                       | 09-24          | 97e2457, 9cd4bc6, 4ec6056 (D3)            | Instrumented real ModelE reproduces the restart bitwise. Chaos floor: 1-ulp perturbation reaches ~1-3% of variability in 5 days, so acceptance beyond one step must be statistical.                    |
| C Track B: components vs real Fortran           | 09-24 to 10-06 | D4-D145; D29, D36, D51, D55, D121-D123    | Dump-hook-and-validate found real bugs (3 in DYNSI, OMEGA constant, 2 land errors D135/D136) and dead code (OCNDYN.f legacy driver, OCNTDMIX.f, OCNQUS.f).                                             |
| D Validation rungs F1 / F2 / F3                 | 10-06 to 10-07 | D127-D129, D149-D157, D163, D165          | Single-step gate needs libimf (3-5% of cloud columns flip without it). One-day runs sit inside the real model's own spread except condensate tails. Radiation can be served by real RADIA (D152-D162). |
| E Reconciliation + owner decisions              | 10-07          | 9cf08e6, 721d789, b71eb22, bd9b16c        | Fidelity work was ahead of the JAX deliverable; criteria were fixed BEFORE the coupled step was compared; C1 and C2 reported separately.                                                               |
| F Hybrid coupled step, stages S0-S8             | 10-07 to 10-08 | D180-D197                                 | Assembled device-resident step bitwise to NumPy (C1), C2 at step 0 MET/PARTLY MET; the 54-step day is NOT MET; every day diagnosis found a cause, none a port defect in clouds.                        |
| G GPU tooling                                   | 10-07 to 10-08 | 5c5b7a4, c7c42c8, d320221, ec36779 (D198) | Environment works on an A100; small sequential ops are launch-bound; no bitwise on GPU; no GPU run of the step yet.                                                                                    |

# Categories A / B / C / D, the gate, and two comparisons

Per prognostic field, maximum absolute difference as a fraction of the field's scale (ACCEPTANCE s3; categories fixed by the project plan and not loosened).

|                        |                                                 |                               |                       |
|------------------------|-------------------------------------------------|-------------------------------|-----------------------|
| <b>A</b> bitwise equal | <b>B</b> $\leq 1e-12$ of scale (rounding level) | <b>C</b> $\leq 1e-6$ of scale | <b>D</b> worse than C |
|------------------------|-------------------------------------------------|-------------------------------|-----------------------|

## Verdict for one coupled step

- MET: every prognostic field in A or B, except named exception columns (at most 10 per field, each  $\leq 1e-9$  of scale).
- PARTLY MET: runs end to end with named exceptions in B or C and the first failing stage identified.
- NOT MET: otherwise. Reported per date (nov26, dec01, jan01), never pooled.
- Bitwise (A) needs the Intel libimf runtime; without it results are B or C and say so.

## C1 versus C2 (reported separately)

- C1, port consistency: JAX-driven step vs our NumPy chained step, same start state, cores, flags, no compile cache. A pass says JAX reproduces our NumPy port, NOT that it reproduces ROCKE-3D.
- C2, fidelity: JAX-driven step vs the real Fortran step-boundary dumps. Only C2 supports a statement about ROCKE-3D.
- Hazards D139, D145, D175: results differ across core counts and with a warm compile cache.

## Multi-day rung (F2, statistical), ACCEPTANCE s4 and s9

- Reference spread: five real one-ulp members of the nov26 day (D151). Classes per field and step: within ( $\leq$  largest member distance), near ( $\leq 2x$ ), beyond ( $> 2x$ ).
- Owner decision 2026-10-08 (s9): the pass rule is 'never beyond 2x the largest member distance, at every step, for every scored field' over ALL steps; 'within at every step' is a secondary line.
- Longer windows: the D172 leave-one-out scoring with the 8 JAN1950 members (D165); the earlier ' $\geq 95\%$  of zonal bins within 2 sigma' rule is not used (true members fail it).

# Identical inputs: what 'the same run' means, and what is compared at each horizon

Same restart, boundary data and configuration (P2SAoM40) for the Fortran and the JAX port; how exact the match can be depends on how far you run.

| Horizon                  | What is compared                                                                                                                    | What 'match' means                                                                                                                                | Status (2026-10-08)                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| One step                 | Full model state after the step: T, U, V, Q, P, cloud water and ice, moment arrays; ocean, ice, lake and land state, field by field | Categories A bitwise, B $\leq 1e-12$ of scale, C $\leq 1e-6$ , D worse; met if every gate field is A or B except named columns                    | nov26, dec01, jan01 MET at step 0 after D199 (nov26 with named column (25,16))          |
| One model day (54 steps) | The same fields at every step, as distance from the real run                                                                        | Statistical: distance $\leq 2x$ the largest spread among five real one-ulp members at every step (ACCEPTANCE s9); bitwise is not expected (chaos) | nov26 NOT MET (D195, QCL step 1 = 2.84); rerun on the D199 land code in progress (D200) |
| Months and longer        | Climate statistics: zonal means, variability, energy balance                                                                        | Not defined yet; 8-member JAN1950 ensemble and D172 leave-one-out scoring exist, no long run exists                                               | not started                                                                             |

## Identical inputs: what that includes today

- Same restart state, boundary data, orbital and configuration settings, same physics code version: our tests start from the real model's own state at a step boundary.
- The model is chaotic: two correct runs drift apart, so beyond a few steps agreement can only be statistical.
- Bitwise matching needs the Intel libimf math library (host callback); on the GPU even exp/pow differ by 1 ulp in 6-12% of values.

## Not yet an independent same-parameters run

- Recorded inputs: vegetation (Ent) exports, land forcing and some surface columns are taken from the real model's records each step, so the run is not free-running.
- Radiation is replayed from the record or called through the original Fortran; SOCRATES is not ported.
- The daily lake update at the day boundary is not ported (D196). A real blind comparison needs these computed, and a multi-month criterion defined.

# Single-step (step 0) verdicts over time: honest status per date

dec01 and jan01 were PARTLY MET in every ported-land configuration until D199 fixed the land humidity (step 0, one run each); nothing here is claimed beyond what the ledger says.

| Date / entry    | Configuration                                        | nov26              | dec01          | jan01           | Note                                                                                                            |
|-----------------|------------------------------------------------------|--------------------|----------------|-----------------|-----------------------------------------------------------------------------------------------------------------|
| 10-06 D129      | libimf, RECORDED land patch                          | MET (named cols)   | MET (A/B)      | MET (named col) | nov26 W2GCM 4 cols 1.7e-12; jan01 1 col <= 2.5e-12                                                              |
| 10-06 D129      | libimf, ported GHY                                   | NOT MET            | PARTLY MET     | PARTLY MET      | first failing stage: SURFACE/GHY land cell (62,34)                                                              |
| 10-06 D129      | no libimf (libm)                                     | NOT MET            | NOT MET        | NOT MET         | first failing stage CONDSE: cloud threshold flips in 3-5% of columns                                            |
| 10-06 0ad795b   | libimf, ported GHY after D135/D136 fixes             | MET                | PARTLY MET     | PARTLY MET      | all MET with recorded land                                                                                      |
| 10-07 D187      | HYBRID assembled (libimf callback, replay radiation) | MET, 1 col         | PARTLY MET     | PARTLY MET      | fails ACCEPTANCE items 1 and 3; 44.3 s/step                                                                     |
| 10-08 D191      | assembled device-resident step                       | MET, 1 col (25,16) | PARTLY MET     | PARTLY MET      | worst EGCM 9.9e-9 (dec01), 7.6e-9 (jan01); item 1 holds only under owner decision 8.3 with named exceptions     |
| 10-08 D199      | assembled step after the land qg_aver/elhx fix       | MET, 1 col (25,16) | MET (1/13/0/0) | MET (1/13/0/0)  | worst gate field 4.1e-13 (dec01), 6.3e-13 (jan01); C1 559/559; step 0 only, replayed radiation, libimf callback |
| 10-08 D195 / s9 | 54-step nov26 day vs 5 real members                  | NOT MET            | not run        | not run         | QCL step 1 ratio 2.84; steps >= 3 worst 1.13                                                                    |
| 10-08 D198      | libm-mode GPU runner, CPU check                      | n/a (not fidelity) | -              | -               | vs NumPy libm: 555 A + 3 B of 558; vs libimf CPU: A126 B170 C47 D215 (a property of libm, not of the port)      |

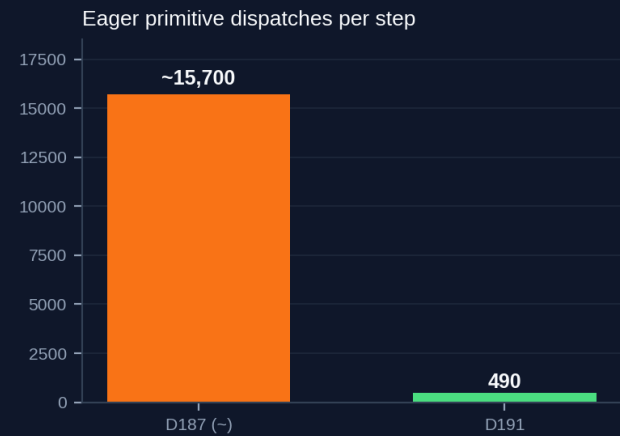
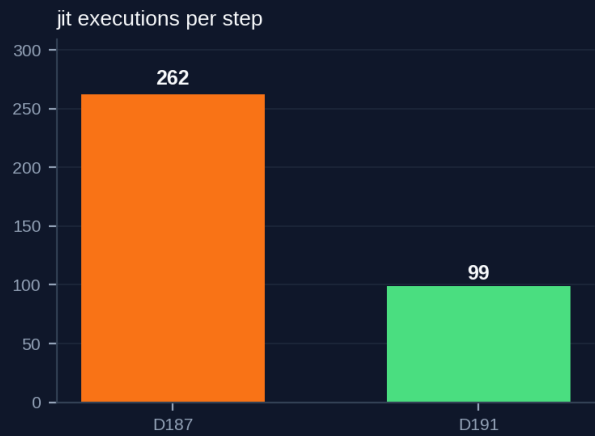
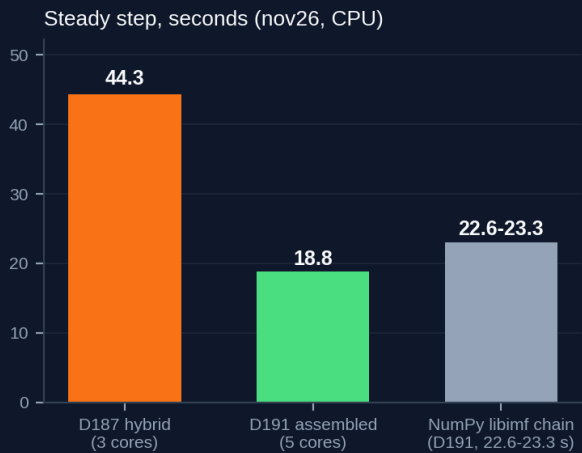
dec01 / jan01 were PARTLY MET because of one land humidity term (D199, fixed at step 0 only; later steps and the day are being re-measured). Four surface-composite exports (QGAVG, USAVG, VSAVG, TGVAVG) were category D against the real end record in D187 and were not re-measured in D191.

# The assembled coupled step (D191) is a hybrid

| Component                        | How it runs in the assembled step                                                                                                                        | Status           | Source                              |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------|
| Atmosphere phase 1 dynamics (J1) | JAX; 75 kernel dispatches from Python; pow through libimf host callback                                                                                  | JAX + callback   | D186, D191                          |
| Clouds / convection              | LSCOND and MSTCNV on device (D189, fused libimf callbacks); QUS subsidence (176 calls, 249.5 MB) and 2 pole columns are host callbacks                   | JAX + callbacks  | D189, D191 s4                       |
| Radiation (SOCRATES)             | Original Fortran RADIA via persistent server (~11 s/call, every 5th step, 33.7 MB to host, 11.6 MB back) or REPLAYED from the real record. Never ported. | Fortran / replay | D159-D162, D185, SOCRATES_PORT_PLAN |
| libimf pow/exp                   | Host callback into the original build's Intel runtime (the fidelity configuration); GPU/libm mode has none                                               | host callback    | ACCEPTANCE s8.1, D183               |
| Surface tiles + land (GHY)       | JAX SURFACE stage on fixed-shape tiles (D188); Ent exports, land forcing and tile radiation columns are RECORDED each step                               | JAX + recorded   | D188, D191 s1                       |
| Post-tile surface + ocean        | Device-resident JAX (GROUND_*, RIVERF, DYNSI, OCEANS, FORM_SI, ADVSI); OADVT2 east-west pre-pass is a host callback                                      | JAX + callback   | D190, D191                          |
| Phase 2 (DISSIP, FILTER)         | jnp, bitwise vs NumPy; pow of SLP/MAtPMB/PEK through libimf callback                                                                                     | JAX + callback   | D191 s1                             |
| Host work every step             | template_build (eager, 294 dispatches), record load (200 calls, 138 MB), day boundary DAILY_ATMDYN in NumPy                                              | host             | D191 s4, D195                       |
| Not implemented                  | new-ice-tile donor rule; daily_LAKE (D196); daily ocean/ice/land updates; transfer-guard test of the full step                                           | missing          | D191 s1, D195 s6, D196              |

D191 verdict on ACCEPTANCE s1: items 2-5 hold as reporting items; item 1 holds under owner decision 8.3 (three jit units plus host calls) for the prognostic state, with named exceptions. 'I do not call it end-to-end JAX.'

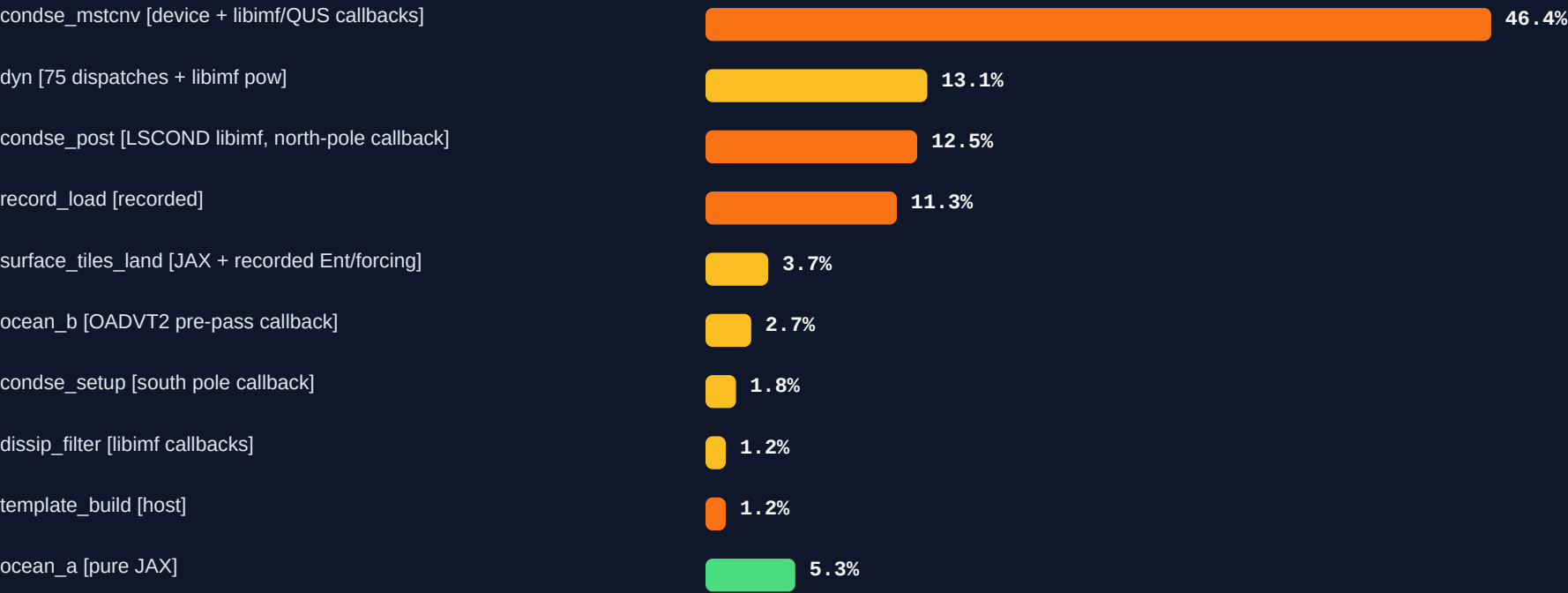
# 44.3 s to 18.8 s per step; 262 to 99 jit executions



| Entry | Change                                                             | Before -> after                                                          | Note                                                     |
|-------|--------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------|
| D182  | cold-compile pathology: XLA flags algsimp + reshape-mover disabled | ~3,000 s -> 390 s cold step 0 (commit 7131447: ~30x faster cold compile) | results bitwise identical                                |
| D188  | SURFACE stage in JAX on fixed-shape tiles                          | ~2x faster, zero eager dispatches inside                                 | category A on 3 dates                                    |
| D189  | MSTCNV as one device program, fused libimf callbacks               | phase 1 of step 0: 33.6 s -> 11.9 s                                      | bitwise equal                                            |
| D190  | post-tile surface + ocean as JAX                                   | 123 -> 5 jit executions; 8.0 s -> 1.76 s                                 | 149/149 category A                                       |
| D191  | assembly of J1 + J2 + J3                                           | steady 44.3 -> 18.8 s; cold first step 349 -> 490 s (worse)              | shared node, one run per number                          |
| D191  | vs NumPy libimf chain, same cores                                  | only 1.2-1.4x faster (22.6-23.3 s)                                       | 46% of step is MSTCNV + callbacks, 13% dynamics dispatch |
| D198  | libm-mode runner on CPU                                            | cold 572 s, steady 11.0 s (load average ~20)                             | indicative; GPU time unknown                             |

# Callbacks and recorded-input work still dominate the assembled step

D191 section 4, nov26, steady, timed run with a device block after every stage (step 17.8 s, stage sum 20.1 s incl. record load).



- Callback bodies (libimf\_ops 5.0 s, fused 0.68 s, QUS 0.78 s, poles 0.39 s, OADVT2 0.26 s = 7.1 s) are included in these shares: about 40% of the step. Radiation: 0.04 s replayed; 12.7 s per call with the real server (1 call per 5 steps).
- Purely-JAX stages (post\_a, post\_b, surface\_pre, template\_apply, tile\_acc, carry\_writeback, melt\_si\_dev, state\_assembly) are each <= 0.2%. The kind labels (JJ/NP/REC) were set by hand from code inspection; the shares are measured.
- Over the 54-step day (D195): steady median 15.9 s (min 15.1, max 18.7); 13 SURFACE rebuild steps of 68-74 s each; cold step 0 428 s; total wall 2,128 s.

# One model day (nov26) vs five real members: NOT MET under ACCEPTANCE s9



| Field | within / near / beyond D203 (D195) | worst ratio steps >= 3 | worst ratio all steps | near steps      | beyond |
|-------|------------------------------------|------------------------|-----------------------|-----------------|--------|
| T     | 54/0/0 (54/0/0)                    | 0.94 (step 41)         | 0.94                  | -               | -      |
| U     | 53/1/0 (53/1/0)                    | 1.03 (5)               | 1.03                  | 5               | -      |
| V     | 54/0/0 (54/0/0)                    | 0.93 (10)              | 0.93                  | -               | -      |
| Q     | 40/14/0 (41/13/0)                  | 1.05 (46)              | 1.05                  | 40-53           | -      |
| P     | 54/0/0 (54/0/0)                    | 1.00 (21)              | 1.00                  | -               | -      |
| QCL   | 50/3/1 (47/6/1)                    | 1.01 (14)              | 2.84 (step 1)         | 2, 13, 14       | step 1 |
| QCI   | 49/5/0 (54/0/0)                    | 1.12 (8)               | 1.12                  | 7, 8, 9, 46, 47 | -      |

Status: NOT MET (owner decision 2026-10-08, ACCEPTANCE s9). The only field beyond 2x is QCL at step 1 (ours 1.15e-6 vs largest member distance 4.06e-7). All 54 steps are finite since the D202 land fix (before it the day was NaN from step 39). 'Within at every step' also fails (U, Q, QCL, QCI near at some steps). The earlier claim of a ~1e-9 floor at steps < 3 was wrong (correction 8a649f7); no exception clause is recorded.

Conditions: HYBRID; radiation REPLAYED (not computed); libimf host callback; Ent, land forcing, template columns recorded; ocean/ice/land daily updates not applied; nit\_strict=False after step 22; one start state; five members. D195 (before the D199/D202 land fixes) in brackets.

# D193-D197: each exceedance traced to a cause

D194 -> D195 (fixed)

## Step 7 divergence

Not the tile set. GHY cells with  $\text{ffnit} > 11$  must regenerate sub-iteration lengths from OUR precipitation; the step used the recorded one. Fix merged (nit\_fix default on): C1 bitwise steps 0-22. The NumPy reference aborts at step 23 (cell (41,21): 11 vs 14 iterations). D193's 'rebuild when the tile set changes' was wrong: the cause is `max_substeps`.

D196 (open, patch not applied)

## Step-48 tile mismatches

54 lake cells ( $\text{FOCEAN} = 0$ ) lose their open-water tile because `daily_LAKE` (LAKES.f:2492) is not applied at the day boundary (FLAKE changes in 632 of 636 lake cells at 47->48). Our NumPy chain has the same omission. Level-2 fix is a ~500-line port, not started.

D197 (explained, still NOT MET)

## QCL step 1, ratio 2.84

One bistable cell (31,12,15) holds 96.8% of the squared error. 1-ulp PK differences from the dynamics exit flip a stratiform threshold; 1 of 8 random 1-ulp draws reproduces it exactly. CLOUDS port from real inputs is category A/B; the NumPy chain has the same value. Without that cell the ratio is 0.51. Does not relax the criterion.

D199 -> D203 (fixed)

## Land runaway, NaN at step 39

D199 (land humidity uses the current substep's `elhx`) fixed `dec01/jan01` at step 0 but sent the `nov26` day to NaN (cell 41,20). D201: the `elhx` switch alone decides finite vs NaN. D202: GHY must receive the PBL's `gusti` (GHY\_DRV.f:1267), not the recorded one. After that all 54 steps are finite; QCL step 1 (2.84x) is unchanged. Mechanism of the original trajectory change is not proven. Correction 8a649f7 (noise floor) stays in force.

# The real model is chaotic; our step is judged against its own spread

## Noise floors measured with the real binary

- D3 (5 days, 1-ulp T perturbation): T layer-1 pointwise rms 0.048 K, global-mean 1.4e-4 K; whole column T rms 0.27 K; only 36 of 257 restart variables stay bitwise identical.
- D151/D149: five real members for the nov26 day; our NumPy and JAX runs diverge from each other exactly like real members.
- D165: 8 real JAN1950 members (ctrl + 7 one-ulp); ctrl reproduces the stored month bitwise;  $t_{500}$  global-mean sd 0.043 K; 8 members = about +/-25% on a std; one season.

## Earlier day results (replayed / free radiation)

- D151 open loop: T, U, V within at all 54 steps; Q near 15 steps, condensates near ( $\leq 1.11x$ ); none beyond 2x.
- D155-D157 free-running radiation (real RADIA server): T/U/V within at every step; condensates up to 1.95x (QCI step 47); radiative flux differences  $< 1$  W/m<sup>2</sup> global mean.
- D171 (Ent computed): QCI worst 1.62; QCL step 1 ratio 2.90 (D192 re-score). The D192 scorer reproduces D157 and D171 exactly.
- None of these days is 'within at every step' for all fields; they were reported before the 2026-10-08 pass rule.

## Consequences

- Single steps are deterministic (floor 0) and can be bitwise only with libimf; beyond step ~1 any rounding difference grows, so acceptance is statistical (ACCEPTANCE s4).
- Without libimf, cloud thresholds flip in 3-5% of columns: a multi-step run drifts about 0.1-0.2 K after 6 steps (GOAL.md; D129).
- Limits: five members, one start state, one day; heavy-tailed condensates (QCI, QCL) are single-cell dominated, so a ratio near 2 is not a significance statement (D192).

# What can silently change a bitwise or statistical result

| Hazard                        | What was found                                                                                                                                                                                                     | Source                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Intel libimf vs glibc pow/exp | Bitwise needs the real build's libimf. With NumPy/glibc dynamics differ at ~1e-13 and cloud columns flip in 3-5%. pow/exp/log are not correctly rounded: ~0.02% of a KPP table differs by 1 ulp (D54).             | README caveats; D54; D129 |
| Single-precision literals     | Fortran literals without d0 (e.g. <code>**(1./3.)</code> in OCNKPP.f) are single precision under the build; <code>REAL*4 DZDH1</code> in RIVERF.                                                                   | D54, D167                 |
| REAL*16 sites                 | Only Ti/Ti2b in SEAICE.f; a binary128 emulation makes seaice_to_atmgrid and ADDICE bitwise; not applied to existing modules.                                                                                       | D173                      |
| XLA CPU flags                 | FMA fusion and x/c rewrites break bitwise; needs <code>--xla_cpu_max_isa=AVX</code> and algsimp (and reshape-mover, D182) disabled before importing JAX; such tests need separate processes.                       | README caveats; D182      |
| Core count / compile cache    | Existing code gives different results on 1 and 3 cores; a warm JAX compile cache changes results. Use identical affinity and no cache for validated runs.                                                          | D139, D145, D175          |
| Host callbacks on one core    | <code>io_callback/pure_callback</code> inside jit DEADLOCK on a single core (0.1% CPU for 23 min). Give such jobs <code>&gt;= 2</code> cores.                                                                      | D185; README              |
| NumPy vs jnp semantics        | <code>x**4</code> is libm pow in NumPy but repeated multiplication in jnp; <code>jnp.cumsum</code> is not a left-to-right sum.                                                                                     | SOCRATES_PORT_PLAN s3     |
| GPU device math               | <code>exp</code> , <code>sin</code> , <code>pow</code> differ from NumPy by 1 ulp in 6.14%, 11.96%, 9.07% of values (log 0.22%): category A is unattainable on GPU unless every call goes through a host callback. | gpu/DISCOVER_RUN.md       |

# The A100 probe says: fuse into few large kernels, and do not expect bitwise

## Measured on Discover (NVIDIA A100-SXM4-40GB, JAX 0.6.1 CUDA)

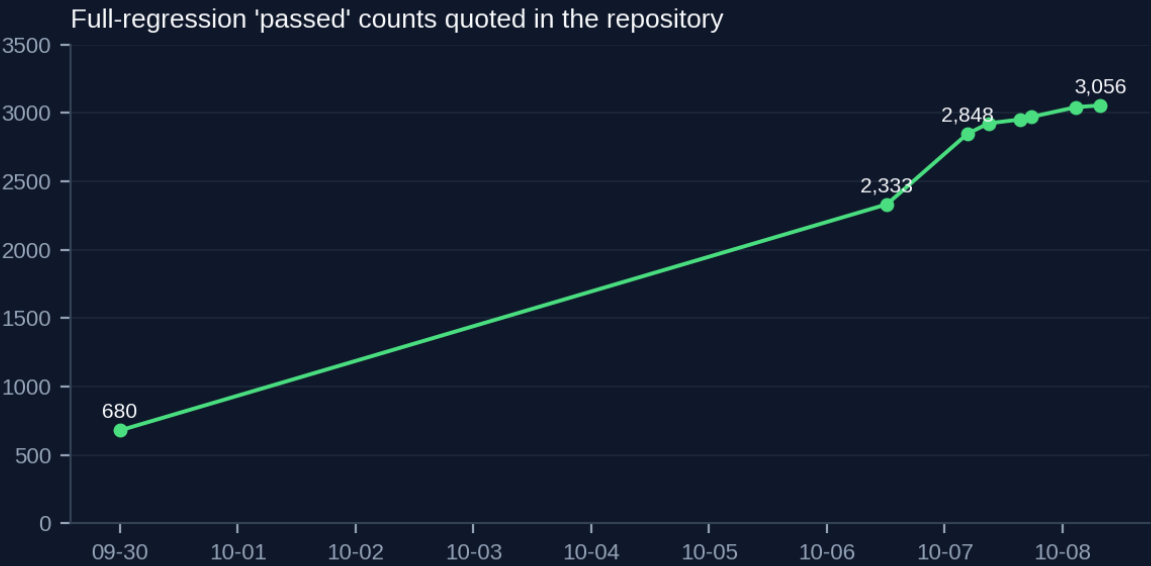
- Smoke test 2026-10-07 (job 58776063): 27 pass, 2 warn (no pytest, no libimf), 0 fail; float64 works; jit+scan on model-sized state compiled in 0.6 s.
- Probe (job 58778982): float64 matmul 2048x2048 1.2 ms (about 820 ms on one CPU core of the work node).
- 100,000 sequential scan steps: 575 ms on the A100 vs 41 ms on one CPU core: launch-bound, ~5.7 us/step; the probe note calls it 14x slower. Long chains of small ops lose; only wide parallel work wins.
- Device exp 6.14%, sin 11.96%, pow(x,0.25) 9.07%, log 0.22% of values differ from NumPy (max 1 ulp).

## What D198 built, and what the first GPU runs show

- gpu\_step\_run.py runs the ASSEMBLED step in libm mode (no libimf), radiation replayed, no NumPy reference needed; data subset 6b (66 files, 1.58 GB), runbook MODE=step.
- On CPU: libm step equals the NumPy libm chain (555 A + 3 B of 558). Against the libimf CPU result: A126 B170 C47 D215: threshold fields flip with ANY different arithmetic (another XLA version: 198 of 558 category D; no CPU flags: 208).
- RUN on the A100 (jobs 58786705, 58788593, 58789387): step 0 finite, within the CPU libimf-vs-libm spread (T 7e-7, Q 2e-4). Cold 726-768 s (655-689 s compile). Steps >= 1: NaN from the surface stage (land/ice/lake/ocean); atmosphere phase 1 is clean; error flags did not catch it.
- A CPU control with the simplifier on is finite; cause on the GPU not known yet (JAX\_DEBUG\_NANS run pending). The 13.1 s step was a repeat of step 0; steps 2-5 took 25 s (NaN steps), the CPU 10-19 s.
- 99 jit executions, ~490 eager dispatches, 176 QUS callbacks per step: launch-bound; the step is not faster on the GPU so far.

**Honest label for any future GPU result: libm mode, radiation replayed, recorded inputs, HYBRID, not bitwise to CPU or Fortran; judged at rounding-level categories for one step and statistically beyond. It must not be used for a C2 / fidelity claim (D198 parent check).**

# 3,056 passed, 6 skipped at a89e117 (plus one known artifact failure)



- Batch regression of a89e117 (clean git-archive export, JOBS=4, cores 8-11, 3,667 s): 3,056 passed, 6 skipped, 1 failed.
- The failure, test\_jax\_harness::test\_header\_complete, asserts a 40-character git HEAD that an export lacks; passes in the real tree. The test was then changed to tolerate a missing .git (commit 1036056); the full suite was NOT re-run after it.
- The 6 skips are gated tests (environment switches) not run in the batch.
- Mutation and non-vacuity checks accompany most ports; flag-sensitive JAX tests run in separate processes (run\_all\_tests.sh; sharded runner 34 min vs ~83 min serial, aggregate totals equal).
- Rule of the owner (2026-10-07): large batches; no full-suite re-run per code change.

Passed counts are what the repository records at each commit (680 after D51 per Project\_Summary; 2,333; 2,848; 2,925; 2,953; 2,972; 3,043; 3,056). They count tests, not validated physics: acceptance is the ledger's per-field categories and the scorer of D192. The 2,471 + 206 entry of 264283d is left out (ambiguous).

# Open work, grouped by what it blocks

## Blocks the day criterion

- Day boundary: apply daily\_LAKE (D196, ~500-line level-2 port, not started); other daily updates (ocean/ice/land) not applied.
- New-ice-tile donor rule (D190 s6): not implemented; multi-day runs without per-step records need it.
- NumPy reference aborts at step 23: C1 beyond step 22 unavailable; nit\_strict policy beyond step 22.
- Leave-one-out scoring with the 8 JAN1950 members (D172) for longer windows: not implemented for the coupled step.

## Blocks 'end-to-end JAX'

- Radiation stays Fortran (served/replayed): SOCRATES port is planning only, 34,899 reached lines, 135-295 h estimate, rule unchanged.
- Recorded inputs: Ent exports (stage-1 port exists, not in the assembled step), land forcing, tile radiation columns, CONDSE entry set, PBL profile columns; AG2OG/IG2OG and straits start in ocean chain.
- Host callbacks (libimf, QUS, poles, OADVT2) and host template build; no transfer-guard test of the full step.
- dec01/jan01 PARTLY MET from the ported land surface (not re-diagnosed).

## Blocks the project's purpose (GPU)

- First GPU run of the assembled step (D198 runner ready; MODE=step).
- Fusing the step into few large kernels (probe: launch-bound).
- Speed without the libimf callback was not measured (D191 s5).
- Reference identity: our run is not shown to be the supplement run P2SAoM40\_003 (PROVENANCE\_MANIFEST); the 100-year climatology comparison is a later milestone, not tested.

Effort: no current total is derived. GOAL.md (10-06) says ~100-170 h more to a month comparison; README\_START\_HERE (10-07) says that figure is too high but gives no new total; treat any total as provisional.

# Owner decisions taken (10-07, 10-08) and decisions still pending

## Taken (recorded in the repository)

- 2026-10-07 (GOAL.md): 'port' = end-to-end JAX first, validated components second; success = one coupled step first, then multi-day; a Fortran radiation callback is acceptable but must be labelled; match P2SAoM40 first; one-month and 100-year comparisons are later milestones.
- 2026-10-07: ACCEPTANCE\_CRITERIA approved before any coupled step was compared; later changes must be dated, never silent.
- 2026-10-07 (s8): headline C2 uses a labelled libimf host callback and is also reported in libm mode; three jit units plus host calls satisfy s1 if boundaries and non-JAX stages are reported; D176 files adopted, D178 after verification.
- 2026-10-07: Ent exports stay recorded up to one day; Ent port started. One writer per working tree. Test/push rule: large batches.
- 2026-10-08 (s9): day pass rule = never beyond 2x at every step; nov26 day NOT MET; no exception for steps 0-2; relaxation only later, only to advance to the GPU run, dated, with both statuses.

## Pending for the owner (as the documents stand)

- Whether to relax the day criterion to advance to the GPU run (s9.3): must be a dated decision after the results, stating the unrelaxed status next to it.
- nit\_strict policy beyond step 22 and what to do about the NumPy reference abort at step 23 (D195).
- Whether to implement daily\_LAKE (D196 patch proposal, level 2 ~500 lines) and the new-ice-tile donor rule.
- Name the GPU host formally (ACCEPTANCE s7.4 / s8.5 still list it open; Discover A100 smoke/probe jobs exist) and when to run MODE=step.
- Whether carbon outputs are part of the F3 acceptance; whether to apply the binary128 Ti/Ti2b diff (speed vs bitwise).
- Whether to lift the rule 'SOCRATES is never ported' (plan: 135-295 h, decisions D1-D6).
- With the project lead: the final comparison target (published 100-year climatology vs a shorter run), identity of our reference with P2SAoM40\_003, where ANN4099.aijP2SAoM40.nc lives.

# Where documents disagree, and what this deck did about it

| Topic                            | Disagreement found                                                                                                 | Used here        |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------|
| Ledger range                     | The request mentions D1..D199; the ledger ends at D198 (HEAD ec36779).                                             | D1-D198          |
| Step time                        | GOAL.md (10-06): ~6 s per step plus clouds; D187: 44.3 s; D191: 18.8 s steady.                                     | D191 (newest)    |
| Remaining effort                 | GOAL.md ~100-170 h; README says too high, no new total; older 30-65 h understated.                                 | no total claimed |
| Noise floor at steps < 3         | D192/D193/D195 '~1e-9'; correction 8a649f7 shows 4e-7 (QCL, step 1).                                               | correction       |
| SURFACE rebuild cause            | D193: tile set changes; D195: max_substeps.                                                                        | D195             |
| Day table counts                 | D193 Q 43/11/0, QCL 49/4/1; D195 Q 41/13/0, QCL 47/6/1.                                                            | D195             |
| C1 array count                   | D187 559/559; D191 558/558 (p4_ij host index array is reference-only).                                             | 558 for D191     |
| Summary page age                 | Project_Summary_and_Conclusions.md stops at D51 (680 tests); STATUS.md is Track A.                                 | ledger + README  |
| GPU host                         | ACCEPTANCE lists the GPU host as to be named; README/DISCOVER_RUN record an A100 smoke test and probe on Discover. | both stated      |
| Jit executions in the probe note | DISCOVER_RUN says 'about 125'; D187 262, D191 99.                                                                  | D187/D191        |

Not claimed anywhere in this deck: a pass of the day criterion; end-to-end JAX; any GPU speed-up of the full port; reproduction of the published 100-year climatology (ACCEPTANCE s5); any C2 statement from a C1 result.

# Track A history (superseded) and where everything lives

## Track A (2026-08-28 to 09-24), reduced scope

- Chained GHY/ATURB/SEAICE/LAKES/PBL/SURFACE driver, JAX-vectorized; measured CPU speedups 20-104x per component (Project\_Summary s4A); kernels on GPU 2.3x (DRYCNV) and 6.3x (PBL); fused chain 4.2x GPU vs CPU on an A100 (STATUS.md).
- Deliberately NOT full fidelity: SEAICE, LAKES and part of ATURB were placeholders; GHY found to be one on 2026-09-23. DRYCNV is not in the real P2SAoM40 binary (ATURB is; commit b2e0487).
- The earlier deck (status\_deck\_2026-10-01.pdf, built by build\_pdf\_2026-10-01.py) reported 14 of 17 modules, 2.3-6.3x GPU kernels and a stop-the-port recommendation; the owner reversed that, so those figures are history, not current status.
- Live artifact of 2026-09-24 (deck.json, slides/\*.html) is unchanged and NOT refreshed.

## Sources and rebuild

- Ledger: FULL\_FIDELITY\_DELTAS.md (D1-D198). Index: Reports/README\_START\_HERE.md. Goal and status: Reports/GOAL.md. Criteria: Reports/ACCEPTANCE\_CRITERIA.md (s1, 3, 4, 8, 9).
- Design: JAX\_COVERAGE\_MATRIX.md (stages S0-S8). Plan/provenance: RECONCILIATION\_PLAN.md, PROVENANCE\_MANIFEST.md. Contingency: SOCRATES\_PORT\_PLAN.md. Lessons: LESSONS\_LEARNED.md. Chronology table: Reports/PROJECT\_CHRONOLOGY.md.
- Rebuild: status\_slides/README.md. In short: conda python make\_figures.py DIR (matplotlib); python3 build\_pdf.py status\_deck.pdf --png DIR (system python3, reportlab). Core 11, scratch outside the repository.
- Stage map S0-S8 (JAX\_COVERAGE\_MATRIX s6): S0 harness, S1 state pytree, S2 phase 1, S3 radiation hand-off, S4 surface tiles/land, S5 post-tile + ocean, S6 assembly, S7 one coupled step, S8 multi-day.
- Review by: when the day criterion status changes, a GPU run exists, or the owner records a relaxation.